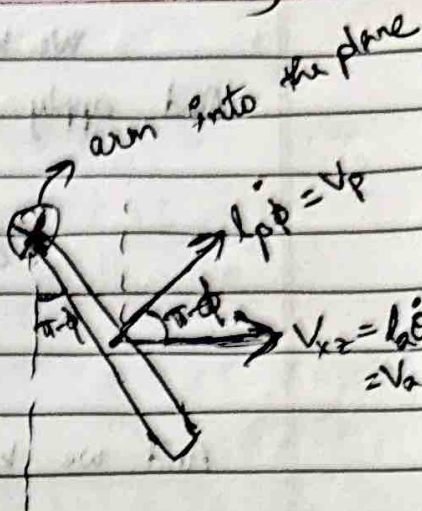
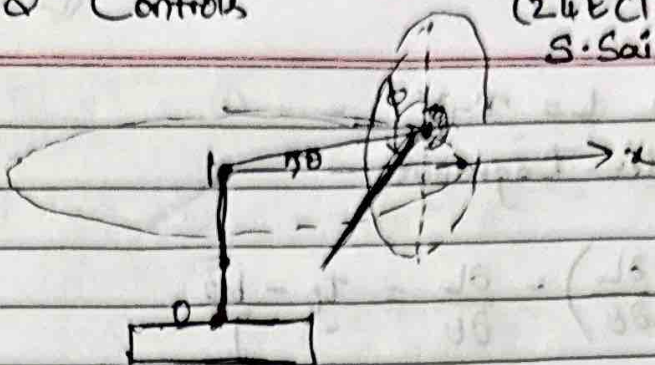




$\theta \rightarrow$ rotary arm angle

$\phi \rightarrow$ pendulum angle

Let state vector $x = \begin{bmatrix} \theta \\ \dot{\theta} \\ \phi \\ \dot{\phi} \end{bmatrix}$ and input torque be T .

In MATLAB, I've took parameters as

$$m_p = 0.2 \text{ kg}$$

$$l_{p/2} = l_p = 0.3 \text{ m} \quad (\text{centre of mass of pendulum rod})$$

$$l_a = 0.2 \text{ m}$$

$$\begin{aligned} (V_{\text{com}})^2_{\text{pendulum}} &= (V_p + V_a)^2 \\ &= (l_a \dot{\theta})^2 + (l_p \dot{\phi})^2 + 2l_a l_p \cos \phi \dot{\theta} \dot{\phi} \end{aligned}$$

Total Kinetic Energy

$$[I_a = m_a l_a^2 / 3; I_p = m_p l_p^2 / 3]$$

standard.

$$T = \frac{1}{2} I_a \dot{\theta}^2 + \frac{1}{2} I_p \dot{\phi}^2 + \frac{1}{2} m_p (V_{\text{com}})^2$$

$$T = \frac{1}{2} I_a \dot{\theta}^2 + \frac{1}{2} I_p \dot{\phi}^2 + \frac{1}{2} m_p [l_a \dot{\theta}]^2 + (l_p \dot{\phi})^2 + 2l_a l_p \cos \phi \dot{\theta} \dot{\phi}$$

$$T = \frac{1}{2} [I_a + m_p l_a^2] \dot{\theta}^2 + \frac{1}{2} [I_p + m_p l_p^2] \dot{\phi}^2 + m_p l_a l_p \cos \phi \dot{\theta} \dot{\phi}$$

Potential Energy $V = m_p l_p g (1 - \cos \phi)$

$$L = T - V$$

We know $L = T - V$

And apply Euler-Lagrange

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = \tau_i - b \dot{\theta}$$

Considering air resistance.

And we know that $\tau_\theta = \tau$ and $\tau_\phi = 0 \rightarrow$ (No external torque on pendulum)

After applying E-L Theorem,

We get

(w.r.t θ) $(J_a + m_p l_a^2) \ddot{\theta} + m_p l_a l_p \cos \phi \ddot{\phi} - m_p l_a l_p \sin \phi (\dot{\phi})^2 = \tau$

(w.r.t ϕ) $(J_p + m_p l_p^2) \ddot{\phi} + m_p l_a l_p \cos \phi \ddot{\theta} - m_p g l_p \sin \phi = 0$

So, we can write

Consider $b_\theta = b_\phi = 0$

$$\begin{bmatrix} J_a + m_p l_a^2 & m_p l_a l_p \cos \phi \\ m_p l_a l_p \cos \phi & J_p + m_p l_p^2 \end{bmatrix} \begin{bmatrix} \ddot{\theta} \\ \ddot{\phi} \end{bmatrix} = \begin{bmatrix} \tau \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \ddot{\theta} \\ \ddot{\phi} \end{bmatrix} = \bar{M}^{-1} \begin{bmatrix} \tau - b_\theta \dot{\theta} \\ -b_\phi \dot{\phi} \end{bmatrix} = \begin{bmatrix} \bar{M}_{11}^{-1} & \bar{M}_{12}^{-1} \\ \bar{M}_{21}^{-1} & \bar{M}_{22}^{-1} \end{bmatrix} \begin{bmatrix} \tau \\ 0 \end{bmatrix}$$

$\Rightarrow$ Linearization Around Upright ($\phi = 0$)

So, $\sin \phi \approx \phi$

$\cos \phi \approx 1$

Linearised Arm Equation:- $(J_a + m_p l_a^2) \ddot{\theta} + m_p l_a l_p \ddot{\phi} = \tau$

Linearised Pendulum Equation:- $(J_p + m_p l_p^2) \ddot{\phi} + m_p l_a l_p \ddot{\theta} - m_p g l_p \phi = 0$

$\bar{M}_{22}$

$\bar{M}_{21} = \bar{M}_{12}$

State Space

$$x = \begin{bmatrix} \theta \\ \dot{\theta} \\ \phi \\ \dot{\phi} \end{bmatrix} \quad \dot{x} = \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \\ \dot{\phi} \\ \ddot{\phi} \end{bmatrix}$$

$$\dot{x} = Ax + Bu$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{M_{12}}{M_{11}} & -\frac{M_{12}}{M_{11}} \\ 0 & 0 & 0 & 1 \\ 0 & \frac{M_{22}}{M_{11}} & \frac{M_{12}}{M_{11}} & -\frac{M_{22}}{M_{11}} \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{M_{11}} \end{bmatrix} \tau$$

$$\dot{x} = Ax + Bu$$

Eigenvalues of A are $0, 0, 7.8302, -7.8302$
 One of them is in RHP
 So, the system is Unstable.

Transfer function

$$M_{11} \ddot{\theta}(s) + M_{12} \ddot{\phi}(s) = \tau(s) \quad \textcircled{*} \quad M_{12} \quad \textcircled{i}$$

$$M_{12} \ddot{\theta}(s) + M_{22} \ddot{\phi}(s) + P\phi(s) = 0 \quad \textcircled{*} \quad M_{11} \quad \textcircled{ii}$$

$$\textcircled{i} - \textcircled{ii} \Rightarrow M_{12} \ddot{\phi}(s) - M_{11} M_{12} \ddot{\phi}(s) - M_{11} P \phi(s) = M_{11} \tau(s)$$

$$\frac{\phi(s)}{\tau(s)} = \frac{M_{12}}{(M_{22} - M_{11} M_{22}) s^2 - M_{11} P}$$

$$\frac{\phi(s)}{\tau(s)} = \frac{-125}{s^2 - 8.882 \times 10^6 s - 61.31}$$

MAKING SYSTEM STABLE

Then, to make the system, i.e. feedback

$$A_{cl} = A - B^*K$$

To get K ,

In matlab, we used

$$Q = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 100 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R = 1$$

$$K = \text{lqr}(A, B, Q, R)$$

We got K matrix = $[-3.1623, -2.3747, -36.47, -7.09]$

Then we get required $A_{cl} = A - BK$

Eigenvalues of A_{cl} =

$$-2.79$$

$$-0.0723$$

$$-0.0292 + 0.0103i$$

$$-0.0292 - 0.0103i$$

All roots are

in LHP

Transfer function for this system

$$G(s) = \frac{\phi(s)}{\tau(s)} = \frac{-125s^2 - 5.92 \times 10^2 s - 1.32 \times 10^4}{s^4 + 1292.7s^3 + 3707.8s^2 + 1.045 \times 10^4 s + 1.93 \times 10^4}$$

After this, we simulated the code for an initial case.